Bentley Oakes

Assistant Professor

Bentley.Oakes@polymtl.ca   Montréal, Canada   bentleyoakes.com   Google Scholar

0000-0001-7558-1434   bentley-oakes   Bentley-Oakes

Key Academic Skills

Researcher: Lead- and co-author of dozens of publications with hundreds of citations on Google Scholar.

Collaborator: Involved in multiple international academic and industrial collaborations.

Teacher/Mentor: Extensive experience teaching courses and mentoring graduate students.

Community-Builder: Organizing and program committee member for local and global events.

Language Skills

English: Native proficiency

French: Conversational (improving)

Education

PhD **McGill University**, Computer Science — Montréal, Canada, Jan 2019
- Thesis: *A Symbolic Execution-Based Approach To Model Transformation Verification using Structural Contracts*
- Supervisors: Hans Vangheluwe and Clark Verbrugge

MSc **McGill University**, Computer Science — Montréal, Canada, Jan 2013
- Thesis: *Practical and Theoretical Issues of Evolving Behaviour Trees for a Turn-Based Game*
- Supervisor: Clark Verbrugge

BSc (Hons Co-op) **University of Manitoba**, Computer Science — Winnipeg, Canada, Jan 2011
- Co-op internship at Electronic Arts Inc., Montréal — Fall 2009, Summer 2010: Prototyping artificial intelligence in commercial video games.
- Co-op internship at Blackberry Limited (RIM), Waterloo — Winter 2009: Implementing cryptographic communication protocols.

Research Experience

Polytechnique Montréal, Assistant Professor — Montréal, Canada, Sept 2023 – present
- Department of Computer and Software Engineering (GIGL)
- Research topics: Accelerating the construction and deployment of digital twins through ontologies for capturing domain knowledge, and large language models and agentic AI for automating their creation. Integrating formal methods to verify digital twin correctness, and assisting users with the configuration and observability of their machine learning and verification/validation services.

Université de Montréal, Post-Doctoral Researcher — Montréal, Canada, Sept 2021 – Aug 2023
- Lab: GEODES Software Engineering Research Group
- Supervisors: Houari Sahraoui and Michalis Famelis
- Research topic: Assisting non-machine learning experts in constructing ML solutions by exploring tailoring of computational workflows.
- Research topic: Extracting and structuring developer rationale from open-source commit messages using NLP and Large Language Models, in collaboration with PhD student Mouna Dhaouadi.

University of Antwerp, Post-Doctoral Researcher — Antwerp, Belgium, Sept 2018 – July 2021
- Labs: Antwerp Systems and Software Modelling, and Constrained Systems Lab
- Supervisors: Hans Vangheluwe and Joachim Denil
- Research topics: Verification and validation of cyber-physical systems, model-driven engineering, multi-paradigm modelling, co-simulation, and digital twins.

Université de Montréal, Visiting Researcher — Montréal, Canada, May 2018
- Host: Eugene Syriani, GEODES Software Engineering Research Group
- Research topic: Developing an interface between the AToMPM modelling tool and the ModelVerse modelling repository.

fortiss GmbH, Visiting Researcher — Munich, Germany, July 2016 – Aug 2016
- Host: Levi Lúcio
- Research topic: Formalizing representations of model transformation languages.

General Motors Technical Center, Visiting Researcher — Warren, USA, Oct 2014 – Dec 2014
- Host: Ramesh Sethu
- Research topics: Applying model transformations for code/model modernization at an industrial scale, and industrial intellectual property concerns.

Industrial Collaborations

Framework for Systematic Design of Digital Twins (DTDesign) — 2019–2021
- Collaborators: Atlas Copco, Flanders Make
- Topics: Digital twin characteristics, integrating knowledge graphs and digital twins.
- Funding: Flanders Innovation and Entrepreneurship Agency (VLAIO)

Automated & Simulation-based Functional Safety Engineering Methodology (aSET) — 2018–2020
- Collaborators: Dana Belgium NV, Siemens Industry Software (Leuven), Flanders Make
- Topics: Verification of safety-critical cyber-physical systems (formal methods, reinforcement learning-based fault injection for hazard discovery, simulation/visualization), DevOps for functional safety.
- Funding: Flanders Innovation and Entrepreneurship Agency (VLAIO)

Innovation in the Development of Electrical Systems For Aeronautics (INES) — 2019–2020
- Collaborators: Boeing Research & Technology Europe (Madrid), Siemens Industry Software (Leuven), Flanders Make
- Topics: Co-simulation configuration, fault injection in co-simulation.
- Funding: Flanders Innovation and Entrepreneurship Agency (VLAIO)

Network for the Engineering of Complex Software-Intensive Systems for Automotive Systems (NECSIS) — 2013–2016
- Collaborators: General Motors of Canada Ltd., Queen's University, University of Antwerp
- Topic: Verification of model transformations.
- Funding: $14M from Automotive Partnership Canada and Natural Sciences and Engineering Research Council of Canada (NSERC)

Teaching Experience

Polytechnique Montréal, Course Instructor — Montréal, Canada, Jan 2024 – present
- Engineering of Digital Twins (LOG8421E) — Winter 2025, Winter 2026
- LOG8371E
- Software Quality Engineering — Fall 2024
- INF6900AE/INF7900A
- Scientific and Technical Communication — Winter 2024, Fall 2024

Polytechnique Montréal, Guest Lecturer — Montréal, Canada, Jan 2022 – Jan 2024

- LOG8704
- XR Software Development — Nov. 2024: *Digital Twins: Introductions and Intersections*
- LOG6953DE
- Model-Driven Software Engineering — Nov. 2022, Nov. 2023: Model-driven engineering, usage and verification of model transformations

McGill University, Course Lecturer/Coordinator — Montréal, Canada, Jan 2015 – Jan 2018

- COMP 202
- Foundations of Programming — Winter 2015, 2017, 2018 (six terms, avg. 189 students)
- Developed and presented lecture material on Java programming for students with no prior programming experience.
- Created assessments and marking guides; coordinated with other instructors and TAs.

University of Antwerp, Teaching Assistant — Antwerp, Belgium, Jan 2019 – Jan 2020

- 2001WETMTR
- Model-Driven Engineering — Fall 2020
- 2001WETMSI
- Modelling of Software-Intensive Systems — Fall 2019

McGill University, Teaching Assistant — Montréal, Canada, Jan 2012 – Jan 2014

- COMP 202
- Foundations of Programming (×2)
- COMP 250
- Introduction to Computer Science (×2)
- COMP 251
- Data Structures and Algorithms (×3)

Student Supervision

Kérian Fiter — PhD Student — 2024–present

- Digital twin federation, trust, and observability

Carlos Pambo — PhD Student — 2024–present

- Recommendations and patterns for digital twin engineering

Adil Lagrou — PhD Student — 2026–present

- Formalization of digital twin component interoperability

Gabrielle Gallant — Master’s Student — 2025–present

- AI-enhanced digital twins for vibration analysis in manufacturing
- Co-supervised with Oguzhan Tuysuz

Invited Talks and Presentations

Guest lecture — Engineering Digital Twins course, McMaster University — Mar. 2026

- Host: Prof. Istvan David
- *What Does Your Digital Twin Do? A Framework and Tooling for Systematic DT Reporting*

Invited talk — SIG LLODIA (Lecture #3) — Jan. 2026

- *Towards Ontology-based Digital Twin Service Construction and Reporting*

Guest lecture — RAISE course, UCLA — Nov. 2025

- Host: Dr. Maged Elaasar
- *What is a ‘Digital Twin’ and How Do I Build One?*

Tutorial — Software Engineering for Machine Learning Applications (SEMLA) Jun. 2025
- *What is a 'Digital Twin' and How Do I Build One?*

Invited talk — EDT.Community May 2025
- *Systematic Digital Twin Reporting*
- Video recording

Invited talk — Next-Generation Cities Institute, Concordia University May 2025
- *Accelerating Digital Twin Research and Engineering*

Conference paper presentation — ICSE 2025, Ottawa May 2025
- *Building Domain-Specific Machine Learning Workflows: A Conceptual Framework for the State-of-the-Practice* (journal-first)

Guest lecture — Engineering Digital Twins course, McMaster University Mar. 2025
- Host: Prof. Istvan David
- *Accelerating Digital Twin Research and Engineering*

Research presentation — KARI Delegation, Polytechnique Montréal Feb. 2025
- Advanced Air Mobility (AAM) challenges and advancements

Invited talk — Software Engineering at Montréal (SEMTL) Jan 2025
- *Recent Work on Systematic Reporting and Construction for Digital Twins*

Guest lecture — LOG8704 XR Software Development, Polytechnique Montréal Nov. 2024
- *Digital Twins: Introductions and Intersections*

Conference paper presentation — MODELS 2024 Sep. 2024
- *Fault localization in DSLTrans model transformations by combining symbolic execution and spectrum-based analysis* (SoSyM journal-first)

Invited talk — onto:Nexus Forum 2024 Jan 2024
- *Accelerating Digital Twin Construction with Ontologies*
- Forum on Ontological Modeling and Analysis

Invited talk — Polytechnique Montréal Jan 2023
- *Bridging the Gap in Verification and Validation of Complex Systems*

Guest lecture — LOG6953DE, Polytechnique Montréal Nov. 2022
- Host: Prof. Mohammad Hamdaqa
- *Model/Graph Transformations: Specification and Verification*

Invited talk — Software Engineering at Montréal (SEMTL) September seminar Sep. 2022
- *Building Domain-Specific Machine Learning Workflows: A Conceptual Framework for the State-of-the-Practice*

Invited talk — Concordia University Jan 2022
- Host: Prof. Essam Mansour
- *Building Domain-Specific Machine Learning Workflows: A Conceptual Framework for the State-of-the-Practice*

Invited talk — Université du Québec à Montréal Jan 2022
- *Capturing and Utilising Domain-Specific Knowledge*

Community Building

Workshop Organizer — onto:Nexus Workshop 2026 (International Workshop on Ontological Modeling and Analysis) Jan 2026

- Co-located with MODELS 2026, Malaga, Spain.
- Co-organizers: Maged Elaasar (Chair), Wahab Hamou-Lhadj, Mohammad Hamdaqa, Eduard Kamburjan

Workshop Organizer — onto:Nexus Workshop 2025 (International Workshop on Ontological Modeling and Analysis) Oct. 2025

- Co-located with MODELS 2025, Grand Rapids, MI. Hybrid event with IEEE proceedings.
- Co-organizers: Maged Elaasar (Chair), Wahab Hamou-Lhadj, Mohammad Hamdaqa

Lead Organizer — Software Engineering at Montréal (SEMTL) Montréal, Canada
Aug 2022 – present

- Regular seminars for the software engineering researchers in Montréal (~40 attendees per meeting, ~30% are professors).
- Coordinating with meeting hosts on content, venue, date, and maintaining website.

Organizing Committee Member 2022–2026

- SEMLA 2026 — General Chair
- FSE 2026 — Student Volunteer Co-Chair
- MODELS 2025 — Student Volunteer Co-Chair
- ICSE 2025 — Posters Co-Chair
- SEMLA 2024, 2025 — Posters Co-Chair
- EDTconf 2024 — Publicity Co-Chair
- ANNSIM 2024 — Proceedings Co-Chair
- ANNSIM 2022, 2023 — Cyber-Physical Systems Track Co-Chair
- MODELS 2022 — Posters Co-Chair

Session Chair 2021–2024

- SEMLA LLM Ops Day — 2024
- Model-Driven Engineering of Digital Twins Workshop (ModDiT) — 2023
- Artificial Intelligence and Model-driven Engineering Workshop (MDEIntelligence) — 2023
- Consortium for Software Engineering Research (CSER) Spring Meeting — 2023
- Model-Driven Engineering and Software Development (MODELSWARD) — 2021

Program Committee Member 2019–2026

- ASE Research Track — 2026
- ASE NIER Track — 2025
- MODELS Artifact Evaluation Track — 2026
- MODELS — 2025
- ANNSIM — 2021–2024, 2026
- DIGITA Workshop (PerCom) — 2026
- International Workshop on Architecting and Engineering Digital Twins (AEDT) — 2025, 2026
- EMMSAD — 2025
- EDTconf — 2024, 2025
- MDE Intelligence — 2022, 2023, 2024
- International Workshop on Models and Evolution — 2022
- ACM Student Research Competition — 2022
- Spring/Summer Simulation Conference — 2019, 2020

Journal Reviewer 2021–2025

- Journal of Systems & Software (JSS) — 2025
- Information and Software Technology (IST) — 2025
- Journal of Software and Systems Modeling (SoSyM) — 2020, 2021, 2023, 2025 (Top 1% Reviewer 2020, 2021; Best Reviewer Award 2025)
- ACM Transactions on Software Engineering and Methodology (TOSEM) — 2024
- SIMULATION — 2023, 2024 (Best Reviewer Award June 2023)
- Science of Computer Programming (SCP) — 2023
- Journal of Computer Languages (JCL) — 2022
- Journal of Object Technology (JOT) — 2022, 2026
- Empirical Software Engineering (EMSE) — 2022
- IEEE Transactions on Automation Science and Engineering (T-ASE) — 2021

Guest Editor for Journal Special Issue Jan 2023

- SIMULATION — *Modeling and Simulation for Software-Intensive Systems: from IoT to Digital Twins*

Jury Member 2023–2026

- Polytechnique Montréal — PhD Thesis Jury President (2026)
- Polytechnique Montréal — M.Sc. Thesis Jury President (2024)
- McGill University — PhD Thesis External Reviewer (2024)
- McGill University — M.Sc. Thesis External Reviewer (2023)

Grant Reviewer 2024–2026

- NSERC Idea to Innovation grant — 2024
- NSERC Discovery grant — 2025, 2026
- Dutch Research Council (NWO) — 2026

Panelist Jan 2023

- ANNSIM PhD Colloquium Panel — 2023

Departmental Service 2025–Present

- Committee Member, Committee Rayonnement (Communications committee), Department of Computer and Software Engineering, Polytechnique Montréal

Scholarships and Awards

SoSyM Best Reviewer Award Jan 2025

- Journal of Software & Systems Modeling

Simulation Journal Best Reviewer Award June 2023

SoSyM Top 1% Reviewer 2020, 2021

- Journal of Software & Systems Modeling

Best Student Paper Award Jan 2019

- SIMULTECH — for the paper *HintCO – Hint-based configuration of co-simulations*

NSERC Postgraduate Scholarship – Doctoral (PGS D) 2015–2016

- Natural Sciences and Engineering Research Council of Canada

Lorne Trottier Science Accelerator Fellowship 2015, 2016

- McGill University

Harold H. Helm Fellowship 2013, 2014

- McGill University

Grad Excellence Award in Computer Science 2012, 2014

- McGill University

Selected Publications

Model-based digital twin engineering: insights, challenges, and future directions May 2026
Philipp Zech, Souvik Barat, Benjamin Nast, **Bentley Oakes**, Judith Michael, Steffen Zschaler, Balbir Barn, Ruth Breu
10.1007/s10270-026-01368-8 (Software and Systems Modeling (SoSyM))

Automated Extraction and Analysis of Developer's Rationale in Open Source Software Jan 2025
Mouna Dhaouadi, **Bentley Oakes**, Michalis Famelis
10.1145/3729383 (Proceedings of the ACM on Software Engineering (FSE))

DTInsight: A Tool for Explicit, Interactive, and Continuous Digital Twin Reporting Jan 2025
Kérian Fiter, Louis Malassigné-Onfroy, **Bentley Oakes**
10.1109/MODELS-C68889.2025.00030 (MODELS Companion (MODELS-C))

Building Domain-Specific Machine Learning Workflows: A Conceptual Framework for the State of the Practice Apr 2024
Bentley Oakes, Michalis Famelis, Houari Sahraoui
10.1145/3638243 (ACM Transactions on Software Engineering and Methodology (TOSEM))

Toward Intelligent Generation of Tailored Graphical Concrete Syntax Jan 2024
Meriem Ben Chaaben, Oussama Ben Sghaier, Mouna Dhaouadi, Nafisa Elrasheed, Ikram Darif, Imen Jaoua, **Bentley Oakes**, Eugene Syriani, Mohammad Hamdaqa
10.1145/3640310.3674085 (MODELS 2024)

Towards Ontological Service-Driven Engineering of Digital Twins Jan 2024
Bentley Oakes, Claudio Gomes, Eduard Kamburjan, Giuseppe Abbiati, Elif Ecem Bas, Sebastian Engelsgaard
10.1145/3652620.3688261 (MODELS Companion (EDTconf) — Best Short Paper Award)

Toward a Systematic Reporting Framework for Digital Twins: A Cooperative Robotics Case Study Jan 2024
Santiago Gil, **Bentley Oakes**, Cláudio Gomes, Mirgita Frasheri, Peter G. Larsen
10.1177/00375497241261406 (SIMULATION)

openCAESAR: Balancing Agility and Rigor in Model-Based Systems Engineering Jan 2023
Maged Elaasar, Nicolas Rouquette, David Wagner, **Bentley Oakes**, Abdelwahab Hamou-Lhadj, Mohammad Hamdaqa
10.1109/MODELS-C59198.2023.00051 (MODELS Companion (MODELS-C))

Machine Learning-Based Fault Injection for Hazard Analysis and Risk Assessment Jan 2021
Bentley Oakes, Mehrdad Moradi, Simon Van Mierlo, Hans Vangheluwe, Joachim Denil
10.1007/978-3-030-83903-1_12 (SAFECOMP 2021 (Computer Safety, Reliability, and Security))

Publications

- See bentleyoakes.com/publications for a full list of dozens of peer-reviewed publications, including journal articles, book chapters, conference papers, workshop papers, and technical reports/theses.